Quiz 2

Two students arrive late for a math final exam with the excuse that their car had a flat tire. Suspicious, the professor says, “each one of you write down on a piece of paper which tire was flat.”

Let’s assume with the professor that the students are lying—not that you would ever lie!—and each independently picking a tire at random. The two random picks form our chance experiment.

What is the sample space Ω? How many outcomes are there in Ω?

A good sample space to use is the set Ω of pairs (i, j), where i and j are whole numbers between 1 and 4. The numbers denote the tires of the car: “1” stands for “front left”, “2” stands for “front right”, “3” stands for “back right”, and “4” stands for “back left”. In a pair (i, j), the number i represents the first student’s pick, and the number j represents the second student’s pick. For example, (2, 3) means that the first student picked “front right” and the second student picked “back right.”

By the multiplication rule, there are (4)(4) = 16 outcomes in Ω.

Write the outcomes in Ω that make up the event A = “the students pick the same tire.”

A = {(1, 1), (2, 2), (3, 3), (4, 4)}

According to the suspicious professor’s probability model, what is the probability P(A) that the students pick the same tire?

The professor imagines that the two students are picking independently at random from among the tires. According to this model, all the outcomes in the sample space are equally likely. We thus have P(A) = 4/16 or 1/4.